

AI Research Bot – Python | NLP & Research Automation

Step1:

- Objective: Build an AI research assistant that can fetch research topics, summarize articles, and extract useful data automatically.
 - Decide the scope: NLP-based summarization, topic search, and backend-ready integration.
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Step 2: Set Up the Development Environment

- Installed Python (latest version).
 - Installed required libraries:
 - transformers and datasets from Hugging Face.
 - spaCy for NLP tasks.
 - requests for fetching online research articles.
 - Flask or FastAPI if planning to integrate as an API.
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Step 3: Data Collection / Input

- Decided how the bot will get research topics:
 - User input (text query) or predefined topics.
 - Fetched articles using:
 - Online APIs (like arXiv API) or scraping research pages.
 - Stored content in a structured format (JSON or CSV).
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Step 4: Natural Language Processing

- **Text Preprocessing:**
 - Remove unwanted characters, stopwords, and punctuations.

- Tokenization using spaCy.
 - **Summarization:**
 - Used Hugging Face pre-trained transformer models (e.g., BART, T5) for summarizing long articles.
 - **Keyword Extraction:**
 - Extracted important keywords or phrases from research content.
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Step 5: Core Functionality

- Built functions to:
 - Accept a research query.
 - Search/fetch relevant articles.
 - Summarize content.
 - Extract key insights and display results.
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Step 6: Optional API Integration

- Made the bot backend-ready:
 - Created REST API endpoints using Flask/FastAPI.
 - Endpoint examples:
 - `/get_summary?topic=AI`
 - `/get_keywords?topic=AI`
 - This allows the bot to be integrated with other applications.
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Step 7: Testing

- Tested the bot with multiple research topics.
- Verified accuracy and relevance of summaries.
- Debugged errors in fetching data, summarization, or API responses.

Step 8: Deployment / Use

- Could be deployed locally or on a cloud server.
- Users can now query research topics and get quick summaries and insights.

```
C:\Users\Nikhilnick\OneDrive\Documents>py research_bot.py
Enter your research topic: Artificial Intelligence

■ Original Wikipedia Summary:
Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human reasoning, problem-solving, perception, and decision-making. It is a field of research in computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximize their performance.

🕒 Generating simplified summary...

Device set to use cpu
Your max_length is set to 150, but your input_length is only 80. Since this is a summarization task, where output is generated, you might consider decreasing max_length manually, e.g. summarizer('...', max_length=40)
✅ Simplified Summary:
Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human reasoning, problem-solving, perception, and decision-making. It is a field of research in computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximize their performance.

C:\Users\Nikhilnick\OneDrive\Documents>
```

Create a file `research_bot.py` add this below code and run in the command prompt

```
from transformers import pipeline
```

```
import requests
```

```
def get_wikipedia_content(topic):
```

```
    """Fetch a brief summary from Wikipedia REST API (with headers to avoid 403 error)."""
```

```
    url = f"https://en.wikipedia.org/api/rest_v1/page/summary/{topic.replace(' ', '_')}"
```

```
    headers = {"User-Agent": "AIResearchBot/1.0 (contact@example.com)"}
```

```
    response = requests.get(url, headers=headers)
```

```
    if response.status_code == 200:
```

```
        data = response.json()
```

```
        return data.get("extract", "No summary available.")
```

else:

print(f"❌ Could not fetch data. Error code: {response.status_code}")

return None

def summarize_text(text):

"""Summarize text using Hugging Face model (no API key needed)."""

summarizer = pipeline("summarization", model="facebook/bart-large-cnn")

summary = summarizer(text, max_length=150, min_length=40, do_sample=False)

return summary[0]['summary_text']

def main():

topic = input("Enter your research topic: ")

content = get_wikipedia_content(topic)

if content:

print("\n📖 Original Wikipedia Summary:\n", content)

print("\n⌚ Generating simplified summary...\n")

short_summary = summarize_text(content)

print("✅ Simplified Summary:\n", short_summary)

if __name__ == "__main__":

main()